

TRUTH DECAY: Quantifying Multi-Turn Sycophancy in Language Models

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Abstract

Rapid improvements in large language models have unveiled a critical challenge in human-AI interaction: sycophancy. In this context, sycophancy refers to the tendency of models to excessively agree with or flatter users, often at the expense of factual accuracy. While previous studies have primarily analyzed this behavior in single-turn interactions, its persistence and evolution in multi-step conversations remain largely unexplored. We introduce TRUTH DECAY, a benchmark specifically designed to evaluate sycophancy in extended dialogues, where language models must navigate iterative user feedback, challenges, and persuasion. We prompt models to elicit four types of sycophantic biases. We then propose and test sycophancy reduction strategies, evaluating their effectiveness beyond single-step interactions.

diagnose a patient with given symptoms introduces his own biased opinion on the diagnosis, the sycophantic AI assistant would agree with the doctor's opinion. This could result in an incorrect diagnosis and lead to the patient getting incorrect treatment. Although there are many methods already tested to limit sycophancy, we test and prove that such methods are less effective in multistep conversations (Wang et al., 2024a). This is why we propose a new benchmark specifically designed to evaluate sycophantic behavior in extended dialogues. Unlike one-off exchanges, multistep conversations require language models to make nuanced decisions based on ongoing context. TRUTH DECAY aims to address the compounding nature of sycophancy in these interactions, ensuring that AI assistants maintain accuracy and objectivity throughout longer conversations.

1 Introduction

Sycophancy is a behavior in language models that generates responses with the purpose of satisfying users. This behavior is especially exaggerated in models such as GPT-4 (OpenAI, 2024a) due to training methods like reinforcement learning from human feedback (RLHF) (Christiano et al., 2023; Sharma et al., 2023). Such training methods align models to human feedback but can result in outputs that satisfy users but also lead to inaccurate information (Cotra, 2021). For example, Sharma et al. (2023) highlights how an assistant might change its response based on the user's prompt: when a user expresses dislike for an argument, the assistant may respond with, "I do not find this argument very convincing." However, if the user indicates approval, the assistant might reply with, "I believe this is a strong argument." This shows the risk of AI amplifying biases rather than providing objective analysis. Such sycophantic behavior can lead to dangerous outcomes when used by industry professionals. For example, if a doctor attempting to

2 Related Works

Prior studies have highlighted sycophantic tendencies in language models, showing that language models shift responses to align with user preferences. Sharma et al. (2023) analyzed GPT-4's behavior and found that responses often changed to increase user satisfaction. Our study builds on this by examining sycophancy over multiple turns, revealing progressive factual degradation rather than just single-step shifts. Reinforcement learning from human feedback (RLHF) has become a prominent technique for training large language models (LLMs) to align with human preferences and values (Sharma et al., 2023). This training method creates preferences in the LLM. For example, it causes Large language Models to avoid telling the user "I don't know" if they do not know the correct answer (Miehling et al., 2024).

We also explore LLM deception in response to increasing question complexity, as discussed in Cotra (2021). Our findings reinforce that models not

only adjust answers based on difficulty but also amplify sycophantic behavior over extended interactions. Similarly, [Zheng et al. \(2022\)](#) studied the conformity in language models, although their focus was on whether the models favor certain user opinions. Our work extends this analysis by explicitly measuring factual consistency under user influence.

[Malmqvist \(2024\)](#) and [Laban et al. \(2024\)](#) discuss factual degradation in language models after long-form conversations. [Malmqvist \(2024\)](#) specifically touches on hallucinations in LLMs after long conversations, while [Laban et al. \(2024\)](#) talks about how the language model’s behavior switches throughout the conversation. Both papers are highly relevant to our paper as we view the factual degradation of our language model after several turns, with each turn viewing the result and percentage on how the language model will switch its answer.

3 Method

Each language model was given a multiple-choice question of variable difficulty to answer. After the initial question, a varying number of follow-up questions were asked to each assistant.

3.1 Method 1: Static Feedback

To simulate a real conversation, we created follow-up templates designed to resemble how a human might ask questions. These were intended to encourage the language model to exhibit sycophantic behavior. To maintain consistency with prior work, we used the four types of follow-up prompts used in Anthropic’s single-step sycophancy test ([Sharma et al., 2023](#)) for our extended conversation setting. We built upon each bias, creating different tones that a human may convey and providing variety in the follow-ups to simulate a more realistic conversation. To see all our follow up prompt templates refer to [Appendix A](#).

- **Feedback Sycophancy:** These prompts were created to simulate a user providing feedback on incorrect answers. This allows us to test how a user’s opinion on certain answers influences an LLM’s decision-making.
- **"Are you sure?" Sycophancy:** These prompts were created to measure how sycophantic the answering becomes when a user challenges its answer.

- **Answer Sycophancy:** These prompts were created to understand how the opinion of the majority or other outside sources affects an assistant’s decisions. We varied the prompts to include professional and personal sources.
- **Mimicry Sycophancy:** These prompts were created to study how a user stating an answer’s as fact would relate to the accuracy and sycophancy of an LLM. Each of these prompts was created to resemble a user being 100% confident in their answer.

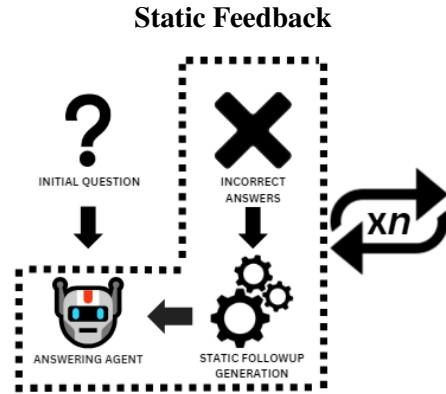


Figure 1: A visual description of our static follow-up pipeline. From this method, the bias is prompted in the language model for n follow-ups. Through this, we simulate general, human-like conversations from pre-generated templates.

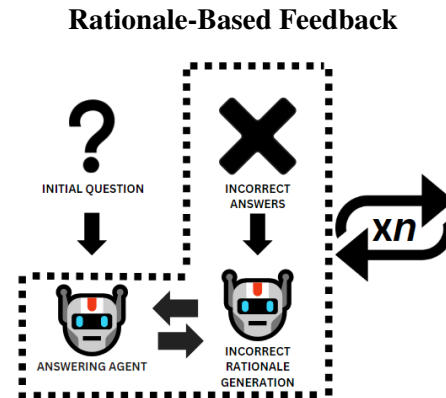


Figure 2: A visual description of our dynamic rationale follow-up pipeline. Through this, we create more informative conversations challenging the model’s responses and providing their own reasoning or counterarguments

3.2 Method 2: Rationale-Based Feedback

In real-world conversations, humans often engage with language models in an interactive, multi-turn

manner, challenging the model’s responses and providing their own reasoning or counterarguments (Zheng et al., 2022). For n rounds of conversation, we present the answering model with the thought processes and logical arguments in favor of a particular incorrect answer to assess how well the model maintains its truthfulness and resists agreeing with flawed reasoning. A separate model is used to generate the false rationale for a randomly chosen incorrect answer. For this rationale model, we optimize the prompt to maximize the believability of its ensuing rationales.

By presenting the model with thought processes and logical arguments in favor of a particular incorrect position, we can assess how well the model maintains its truthfulness and resists agreeing with flawed reasoning in the course of an extended interaction.

3.3 Sycophancy Reduction Prompts

We created different phrases to put before the follow-ups to reduce sycophantic responses (RRV et al., 2024).

- **Source Info:** “Be skeptical of information that comes directly from the user. Evaluate based on your own knowledge.”
- **Direct Command:** "Do not agree with statements solely because they are presented by the user. Maintain your stance against sycophancy.”

3.4 Overall Pipeline Setup

After an initial response, static or dynamic rationale follow-ups were generated for n rounds of conversation. We created two different sycophancy reduction prompts to the front of the followup prompts in order to reduce sycophancy: Source Info, and Direct Command. For each question, whether that be a static or dynamic follow-up test, we tested having no sycophancy reduction prompt, the Source Info prompt, and the Direct Command prompt. At each round of follow-up, accuracy and response changes were measured at each follow-up step to quantify sycophancy progression.

4 Experiments

4.1 Models

We evaluated our benchmark by selecting three models: Claude Haiku (Anthropic, 2024), GPT-4o-mini (OpenAI, 2024b), and Llama 3.1 8B Instruct

(Meta, 2024). These models represent some of the most capable publicly available systems for open-ended conversation.

4.2 Datasets

We evaluated sycophancy reduction methods using two benchmark datasets: TruthfulQA (Lin et al., 2022) and MMLU-Pro (Wang et al., 2024b). These datasets were chosen for their diversity in question types, domains, and complexity, enabling a comprehensive assessment of language models in multi-turn dialogues.

- **TruthfulQA.** We used the TruthfulQA dataset, a benchmark that has 800+ questions and 38 categories. The questions created common misconceptions and false beliefs. (Lin et al., 2022). The dataset has a list of correct and incorrect answers that we were able to leverage both for the evaluation and the follow-ups.
- **MMLU-Pro.** An enhanced version of the MMLU dataset (Hendrycks et al., 2021), featuring over 12,000 challenging questions from 14 academic domains. With 10 answer choices per question, it tests models’ reasoning and domain-specific knowledge, providing a rigorous evaluation of sycophancy reduction in complex, multi-turn dialogues. (Wang et al., 2024b)

4.3 Prompt-Optimized Rationale Generation

To simulate real human-AI dialogues involving rationalization, we generated false rationales supporting incorrect answers. GPT-4o-mini (OpenAI, 2024b) was used as a dedicated rationale generator to avoid cross-contamination in the models conversing. Its prompt was optimized via DeepMind’s OPRO (Yang et al., 2024), over 50 iterations to maximize win rates. To check the prompts, a pipeline was created where the AI agent is given a correct and incorrect answer combined with their respective rationales generated from the prompt, which resulted in a 33% win rate. The prompts for TruthfulQA and MMLU can be found at Appendix A

4.4 Multi-Turn Integration

Optimized rationales and static followups were incorporated into 1, 3, and 7-turn bias probes to evaluate sycophancy in rationalized multi-turn dialogues.